Microsoft

Microsoft Build

May 6–8, 2019



Microsoft

Craft Stunning Mobile Apps with the Xamarin Visual for Design Systems

David Ortinau
Senior Program Manager
Mobile Developer Tools

David Ortinau

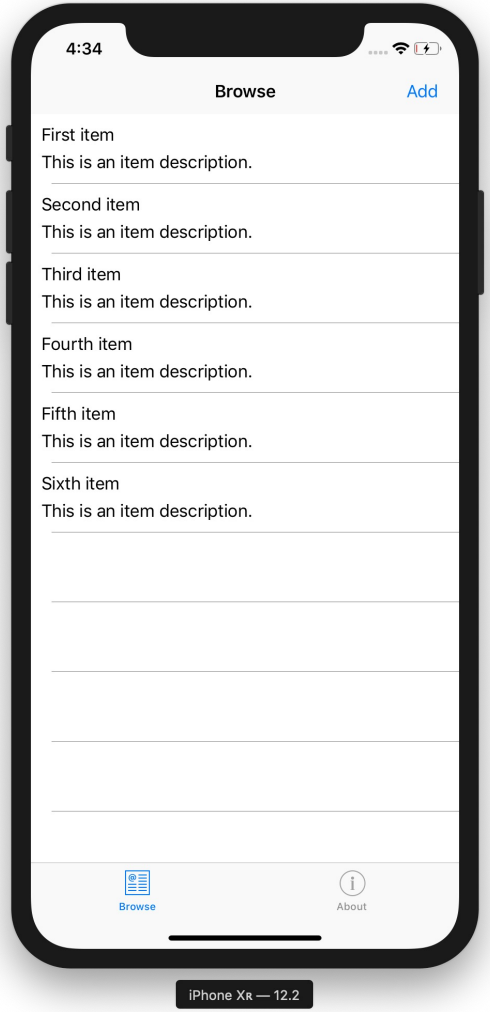
Senior Program Manager
Mobile Developer Tools
Xamarin

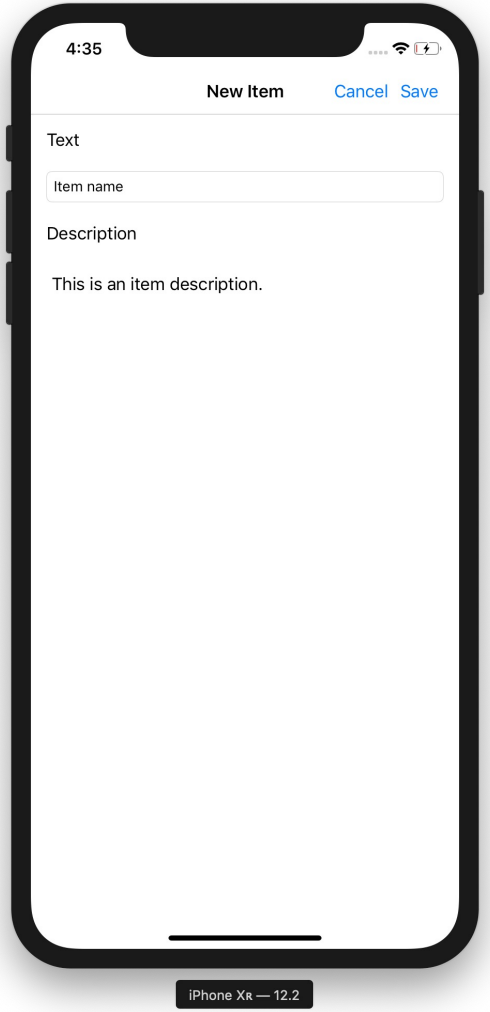
@davidortinau

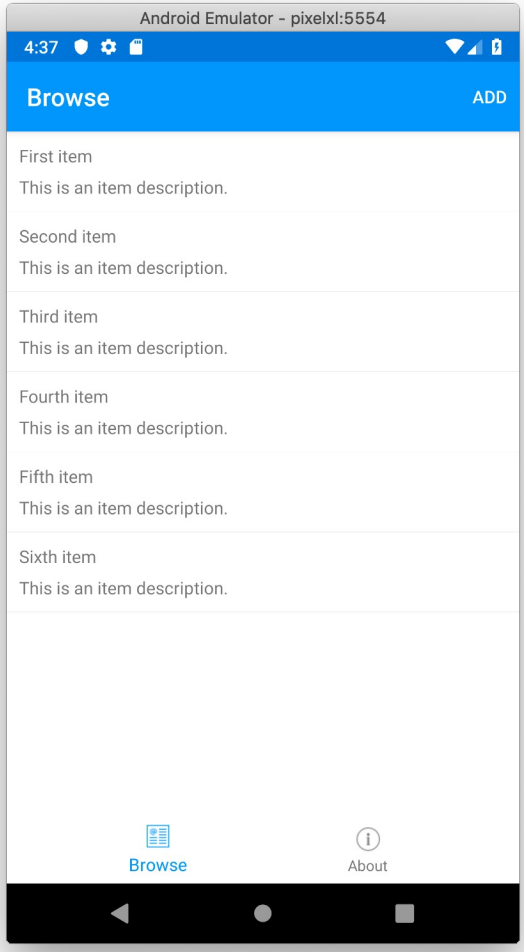
davidortinau

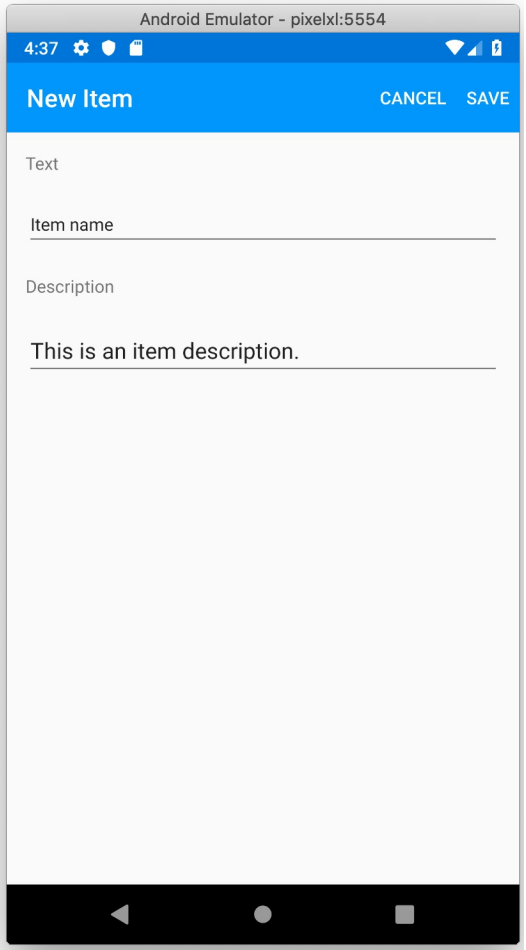


Super Easy to Build Apps Like

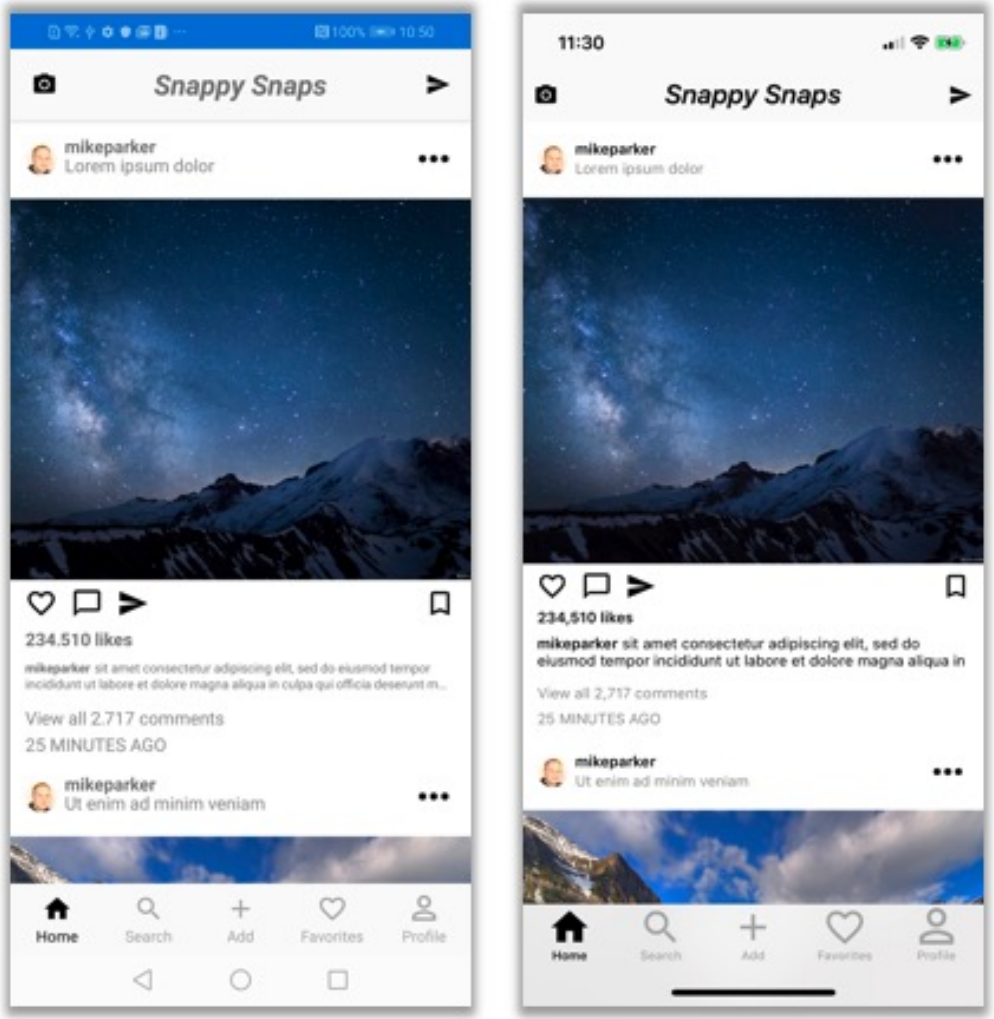




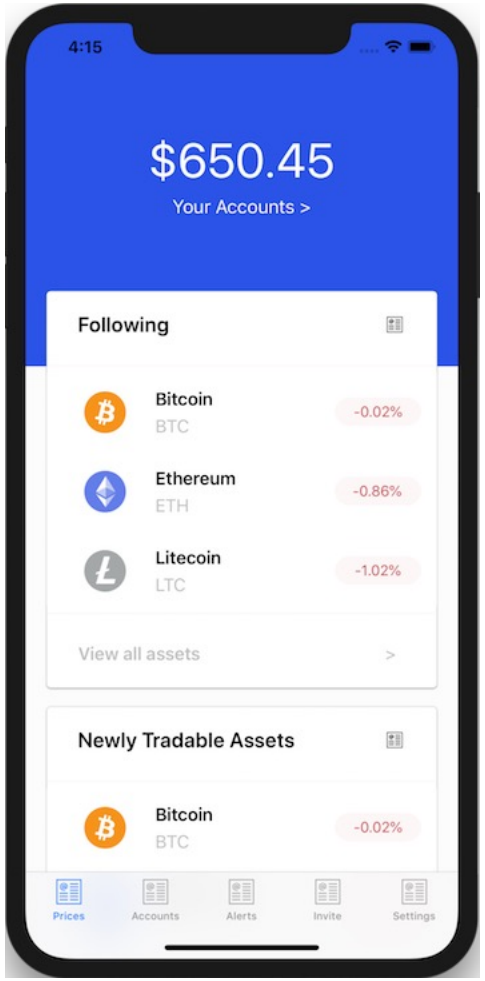


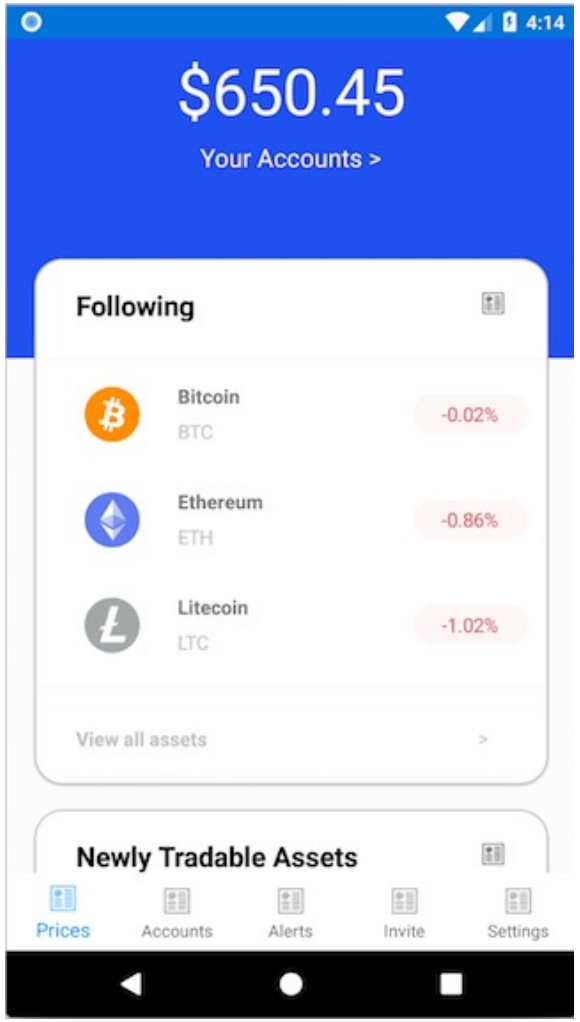


But How Do We Easily Build Apps Like

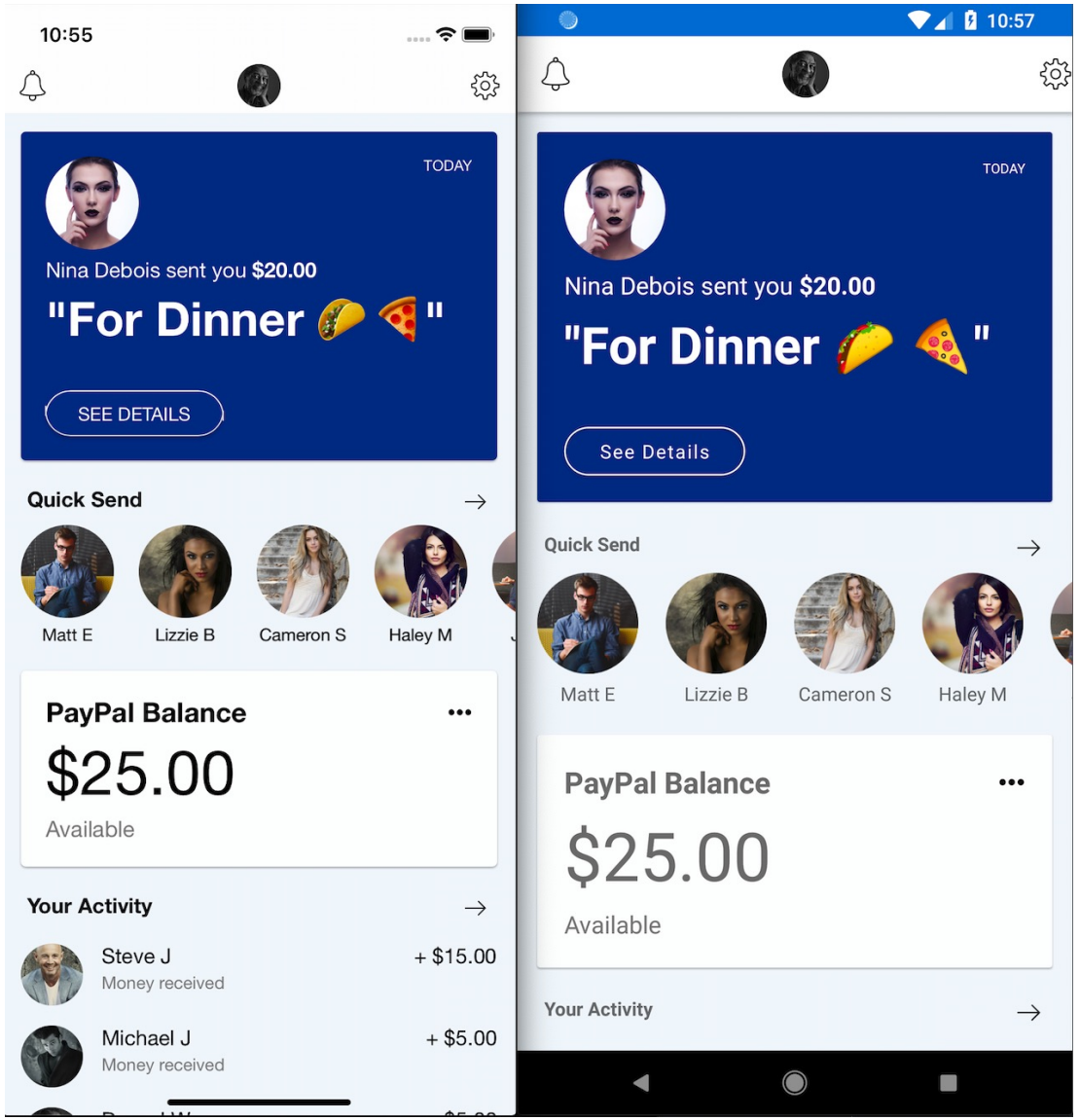


And Like

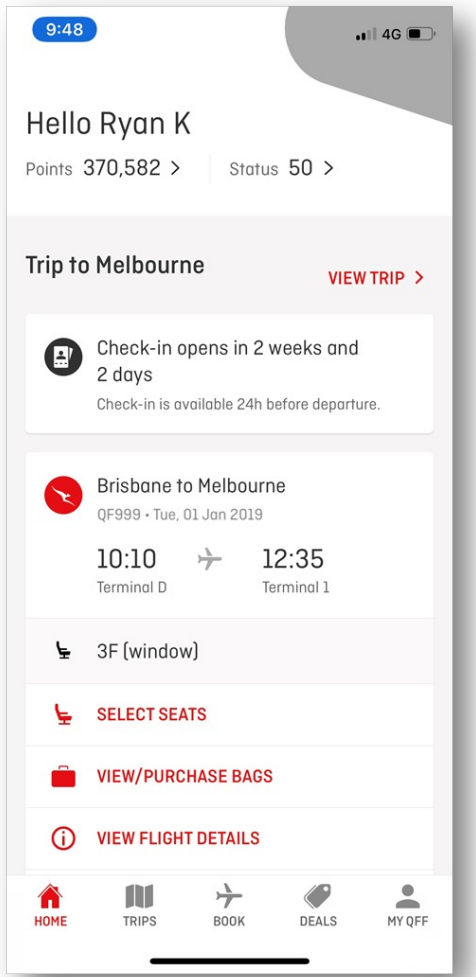

4:15
$650.45
Your Accounts >
Following
Bitcoin
BTC
-0.02%
Ethereum
ETH
-0.86%
Litecoin
LTC
-1.02%
View all assets
>
Newly Tradable Assets
Bitcoin
BTC
-0.02%
Prices
Accounts
Alerts
Invite
Settings


4:14
$650.45
Your Accounts >
Following
Bitcoin
BTC
-0.02%
Ethereum
ETH
-0.86%
Litecoin
LTC
-1.02%
View all assets
>
Newly Tradable Assets
Prices
Accounts
Alerts
Invite
Settings

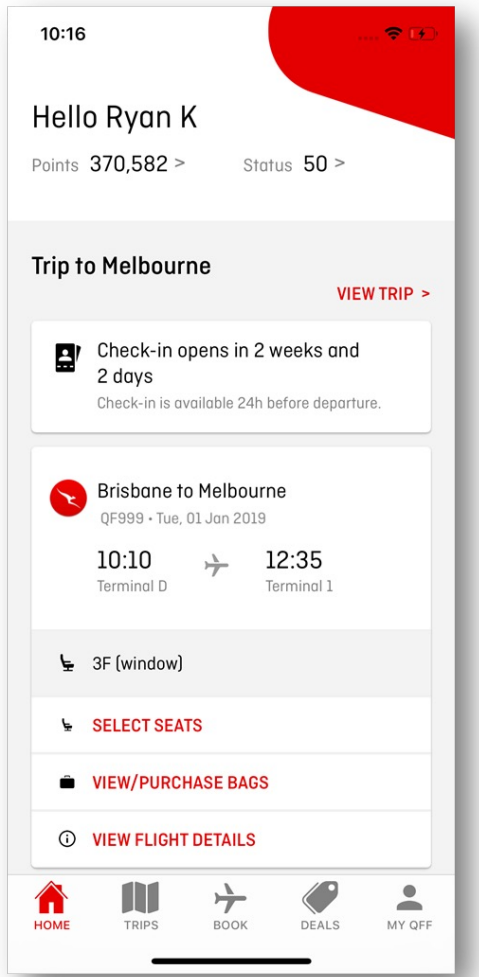
And Like


10:55
TODAY
Nina Debois sent you $20.00
"For Dinner 🌮🍕"
SEE DETAILS
Quick Send
Matt E
Lizzie B
Cameron S
Haley M
PayPal Balance
$25.00
Available
Your Activity
Steve J
Money received
+ $15.00
Michael J
Money received
+ $5.00
10:57
TODAY
Nina Debois sent you $20.00
"For Dinner 🌮🍕"
See Details
Quick Send
Matt E
Lizzie B
Cameron S
Haley M
PayPal Balance
$25.00
Available
Your Activity

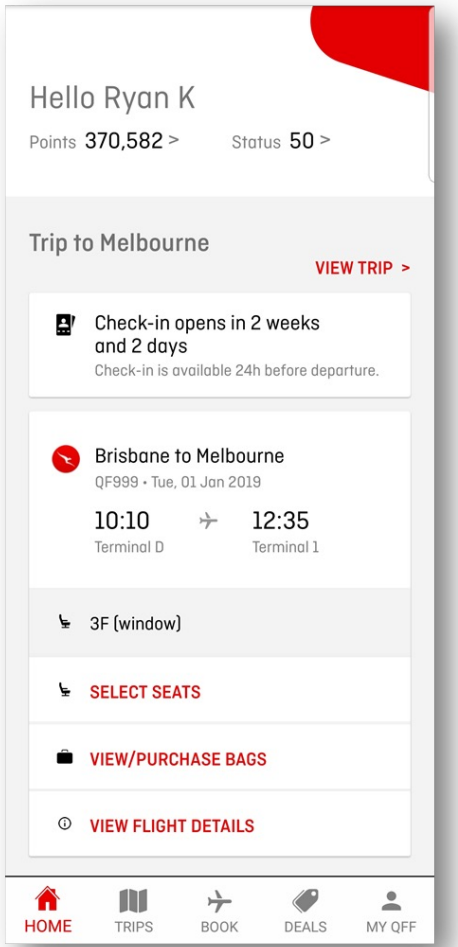
And Like



Source (iOS iPhone X)

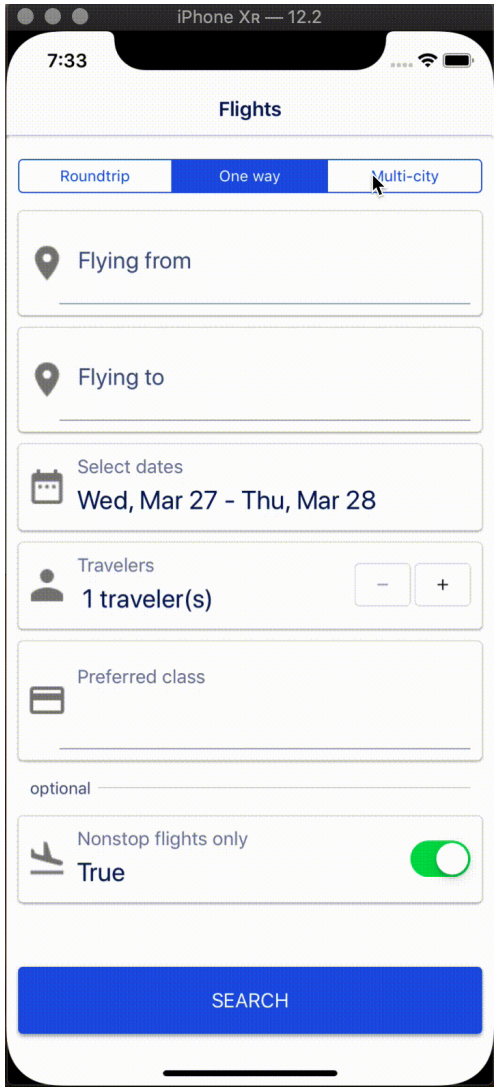


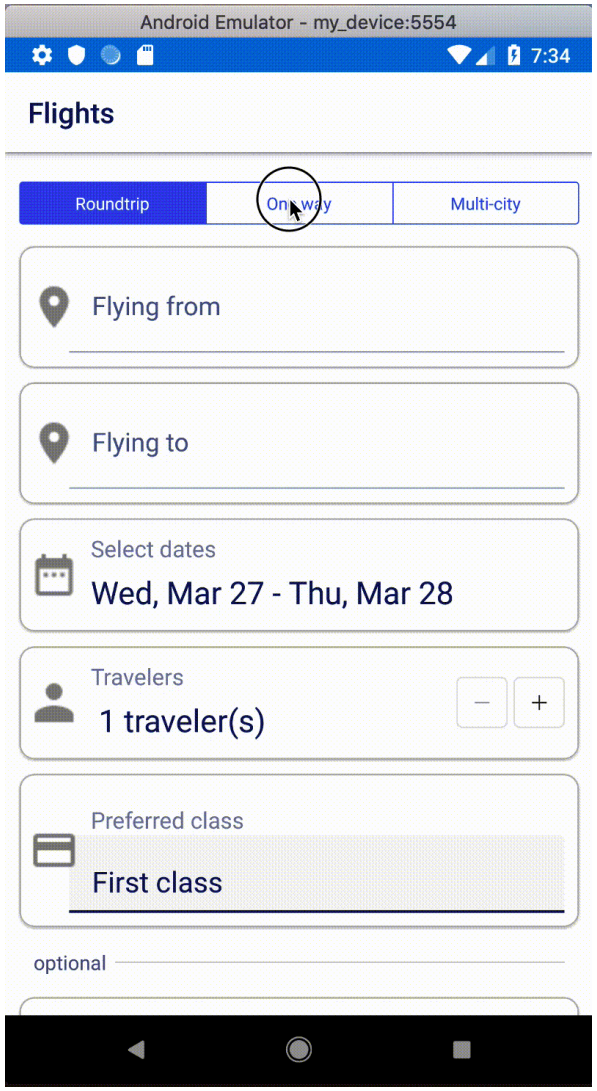
Visual (iOS iPhone X)



Visual (Android S9)

And Like


iPhone XR — 12.2
7:33
Flights
Roundtrip
One way
Multi-city
Flying from
Flying to
Select dates
Wed, Mar 27 - Thu, Mar 28
Travelers
1 traveler(s)
Preferred class
optional
Nonstop flights only
True
SEARCH


Android Emulator - my_device:5554
7:34
Flights
Roundtrip
One way
Multi-city
Flying from
Flying to
Select dates
Wed, Mar 27 - Thu, Mar 28
Travelers
1 traveler(s)
Preferred class
First class
optional


TELL ME!

Visual="Material"



Visual with Material for iOS Components

A unified way to deliver a curated set of renderers to achieve designed looking and feel across platforms.

Available in:

- Xamarin.Forms 3.6
- Xamarin.Forms 4.0-pre10 (Release Candidate)

Visual="Material"

Material Design




MadisonYork



Material Design

- Material is a design system



Material Design

- Material is a design system

Material Design

- Material is a design system

A design system is a series of components that can be reused in different combinations. Design systems allow you to manage design at scale.

https://www.forumone.com/ideas/what-is-design-system/

Material Design

- Material is a design system

Designers wishing to give their suite of products a unique but consistent look and feel define a design language for it, which can describe choices for design aspects such as materials, colour schemes, shapes, patterns, textures, or layouts.

https://en.wikipedia.org/wiki/Design_language

Material Design

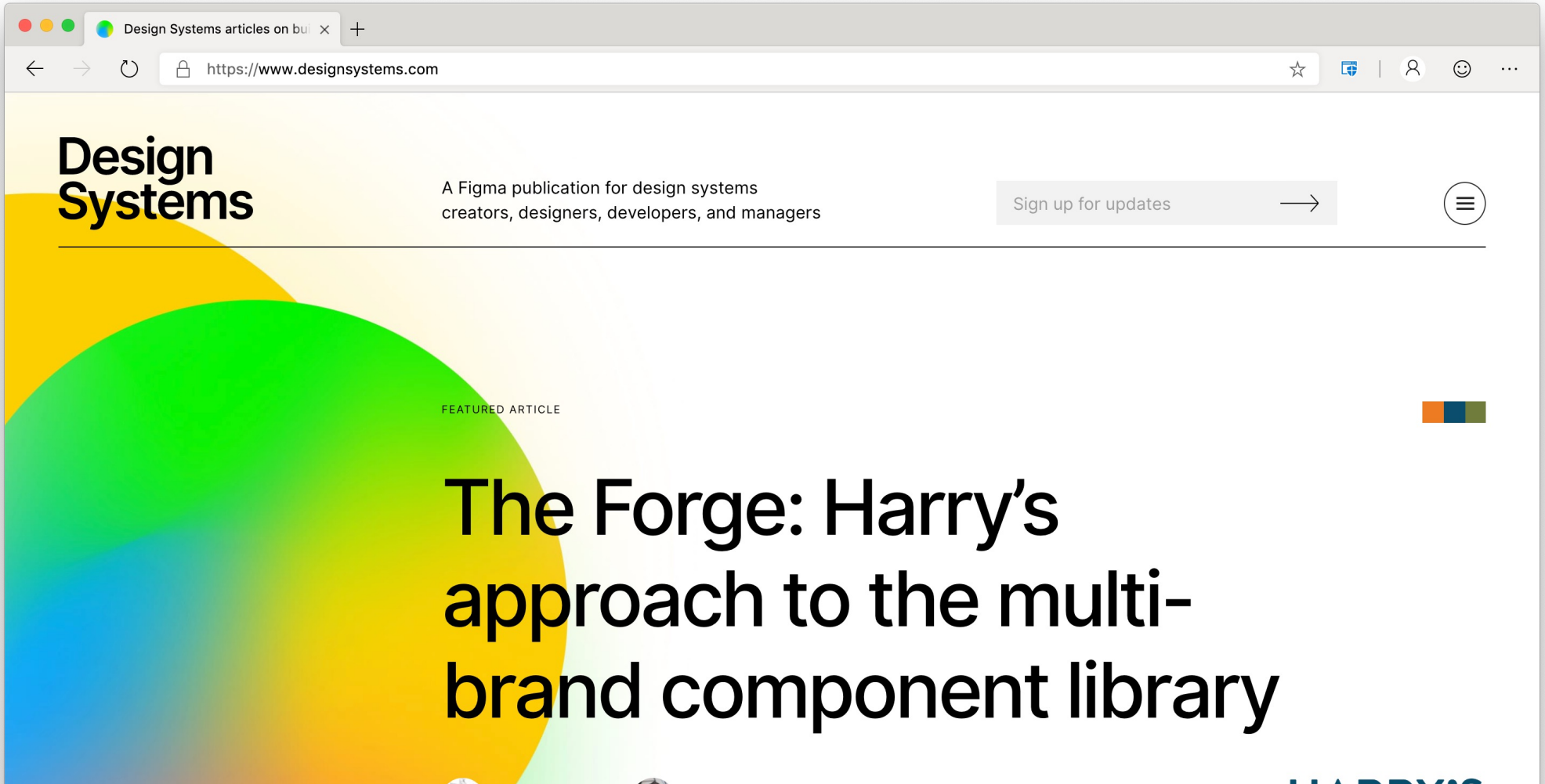
- Material is a design system

Designers wishing to give their suite of products a unique but consistent look and feel define a design language for it, which can describe choices for design aspects such as materials, colour schemes, shapes, patterns, textures, or layouts.

https://en.wikipedia.org/wiki/Design_language

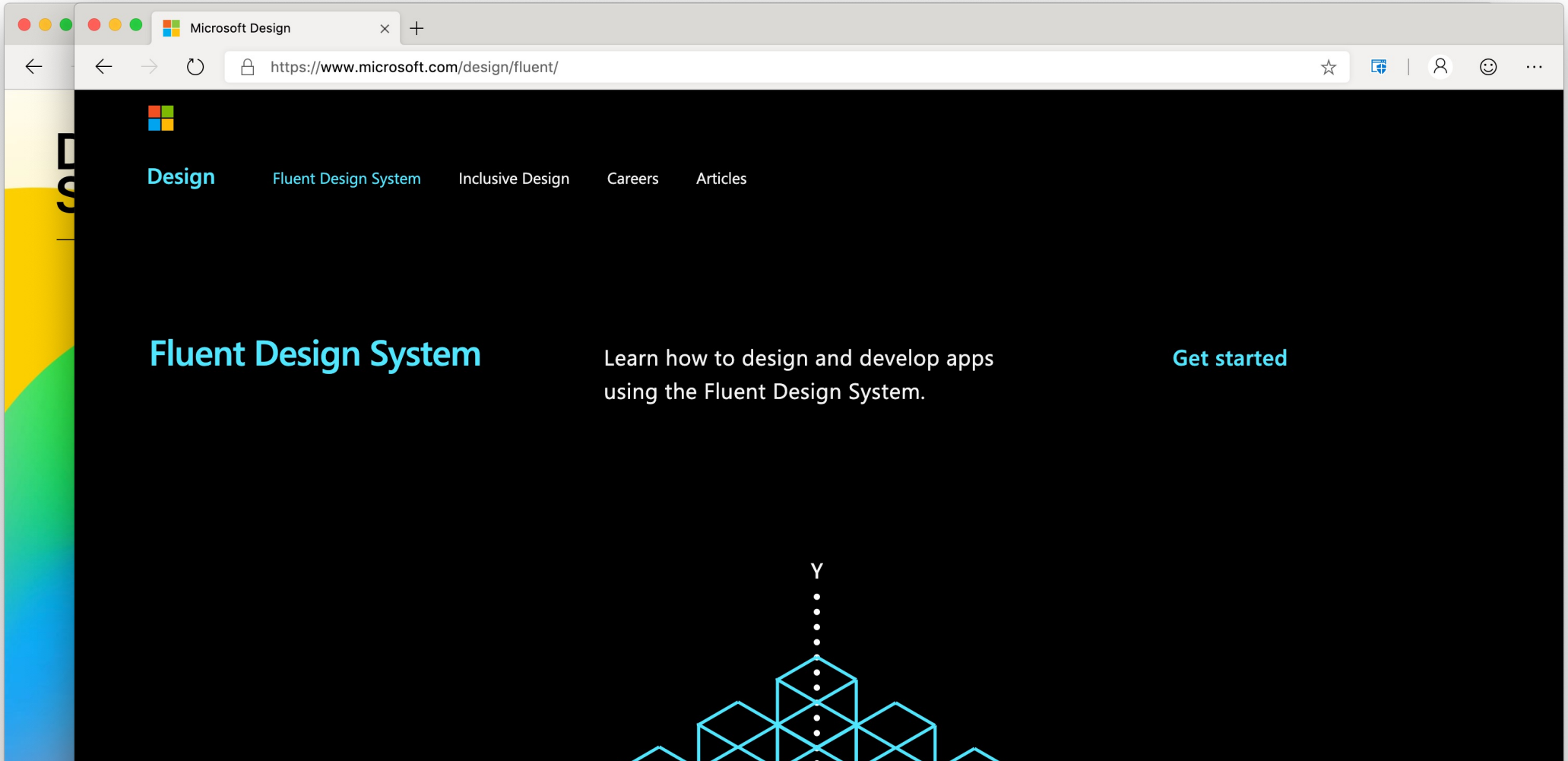
Material Design

- Material is a design system



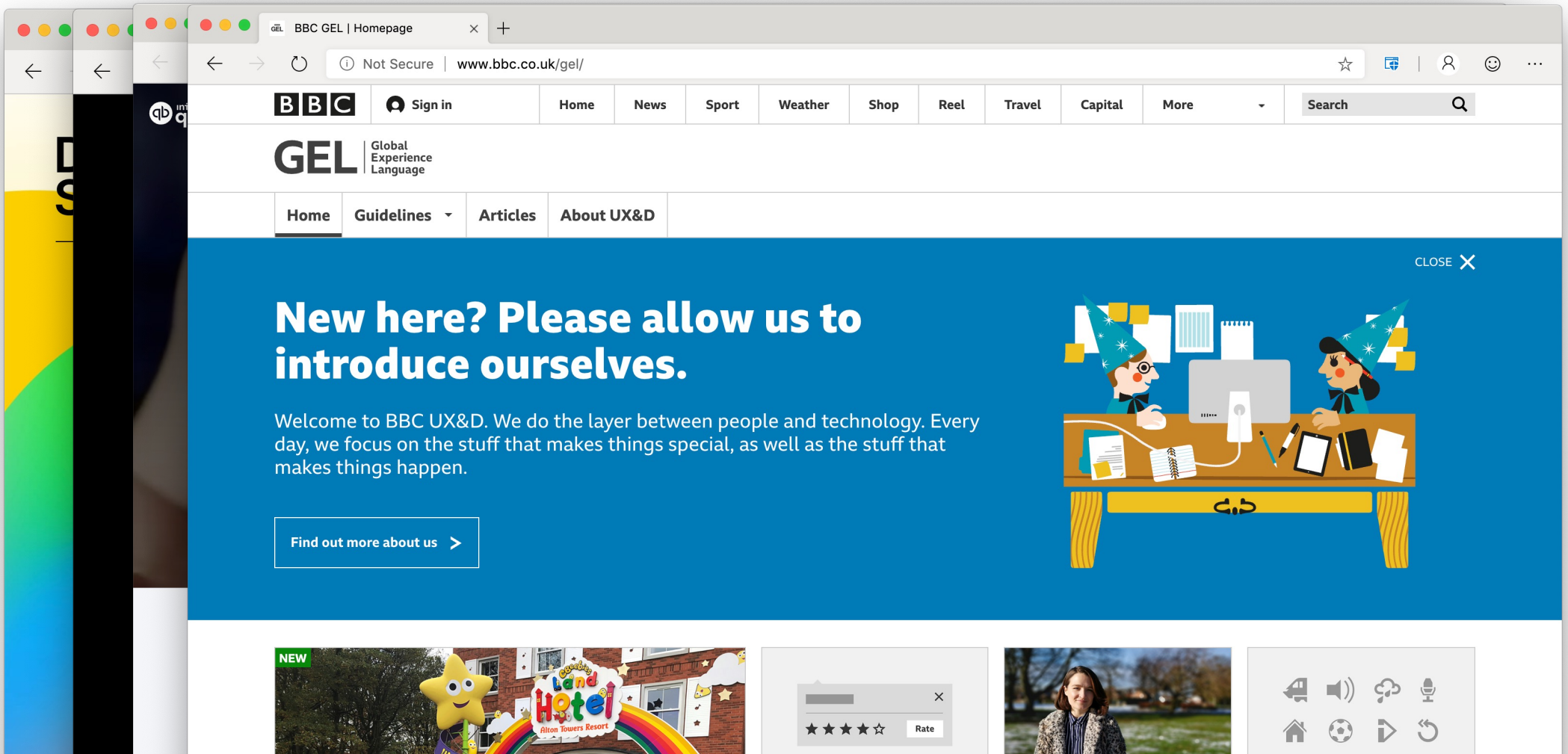
Material Design

- Material is a design system



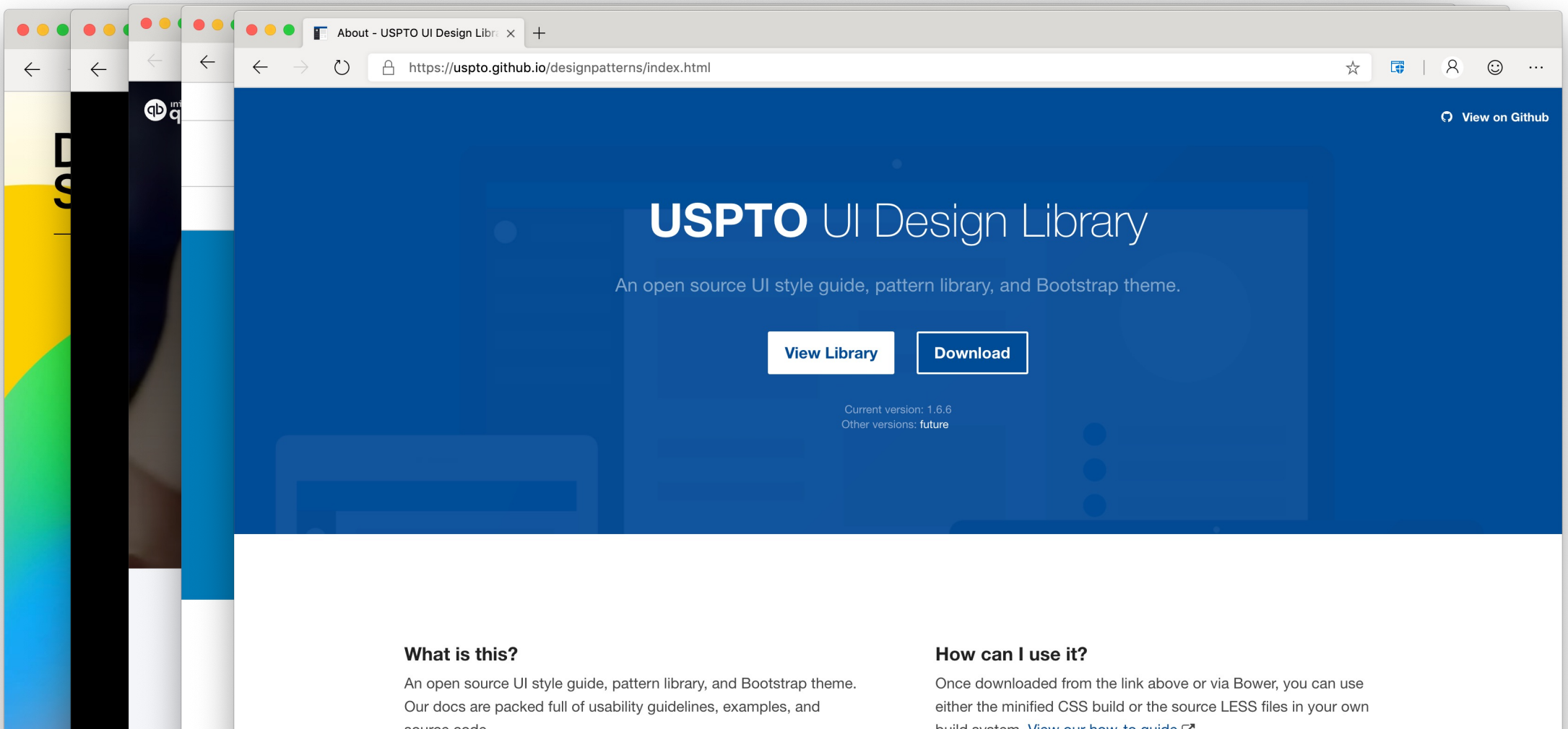
Material Design

- Material is a design system



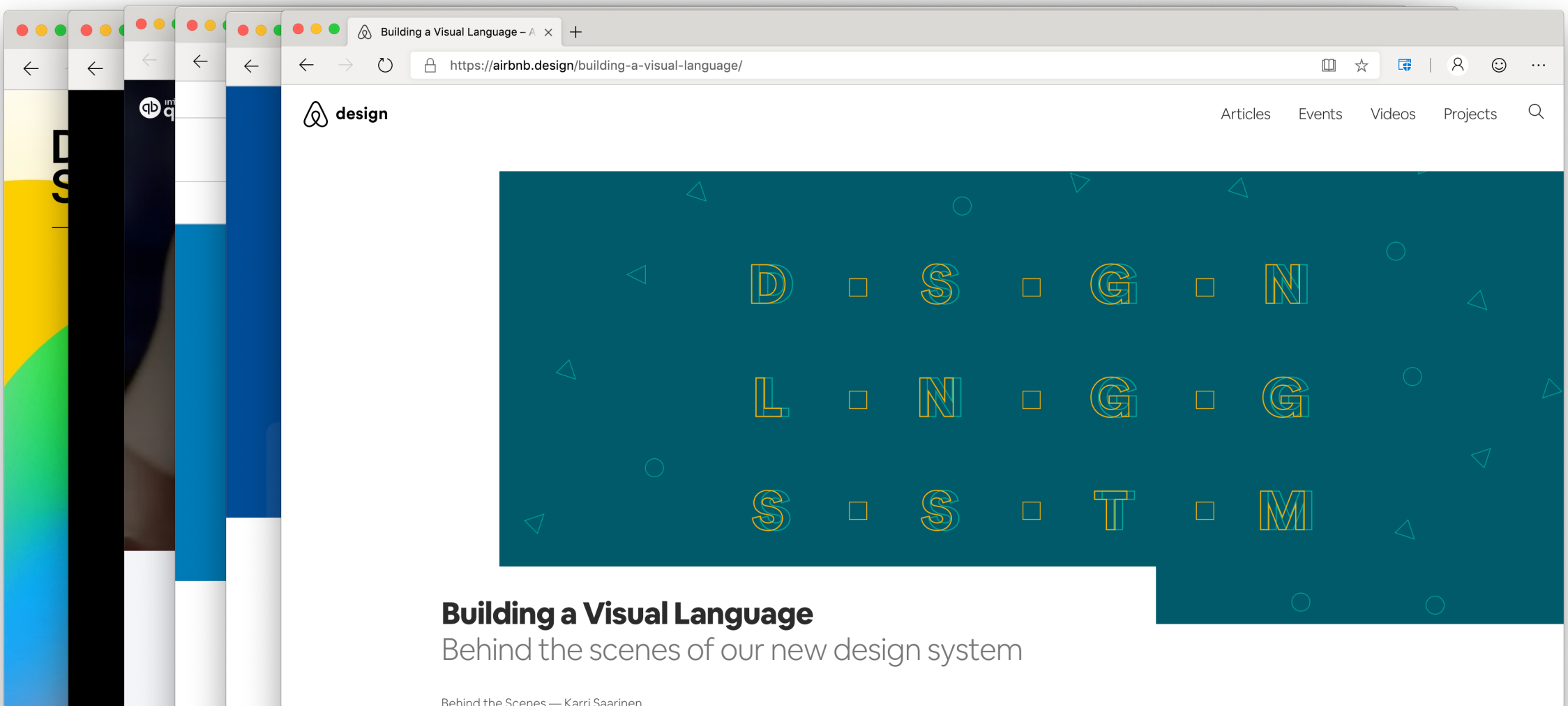
Material Design

- Material is a design system



Material Design

- Material is a design system

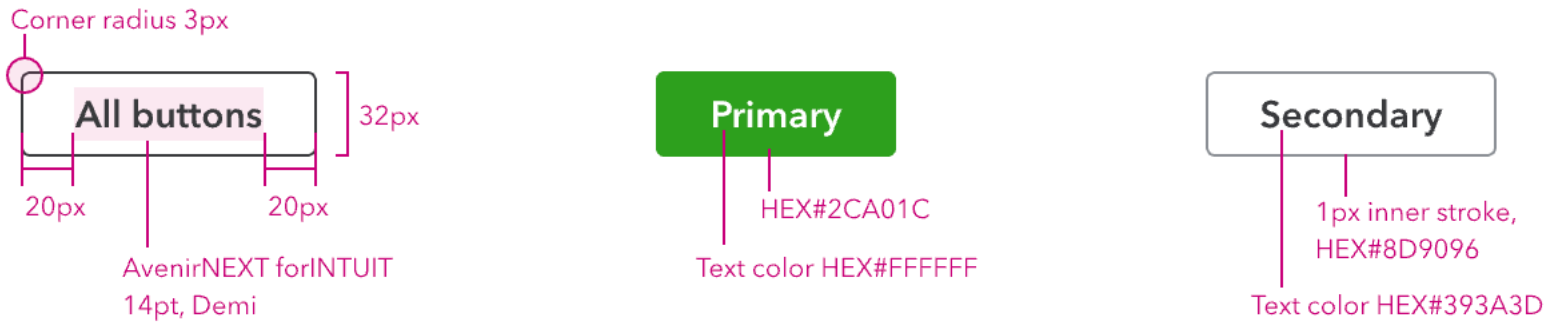




Material Design

- Material is a design system

Basic buttons specs



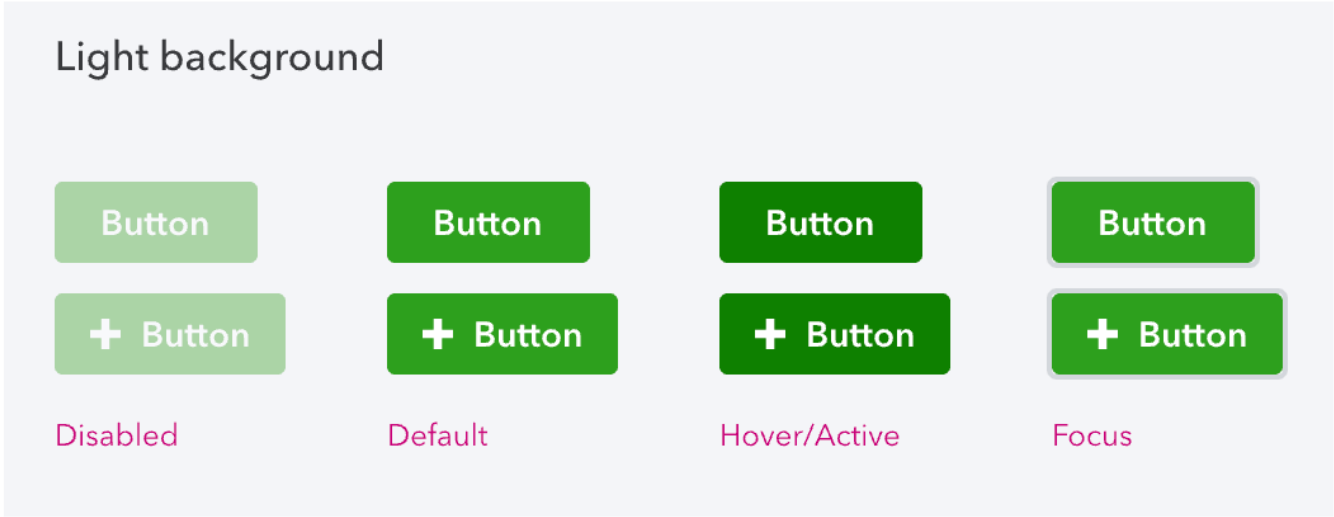
- Height – 32 px
- Border radius – 3 px
- Side padding = 20 px

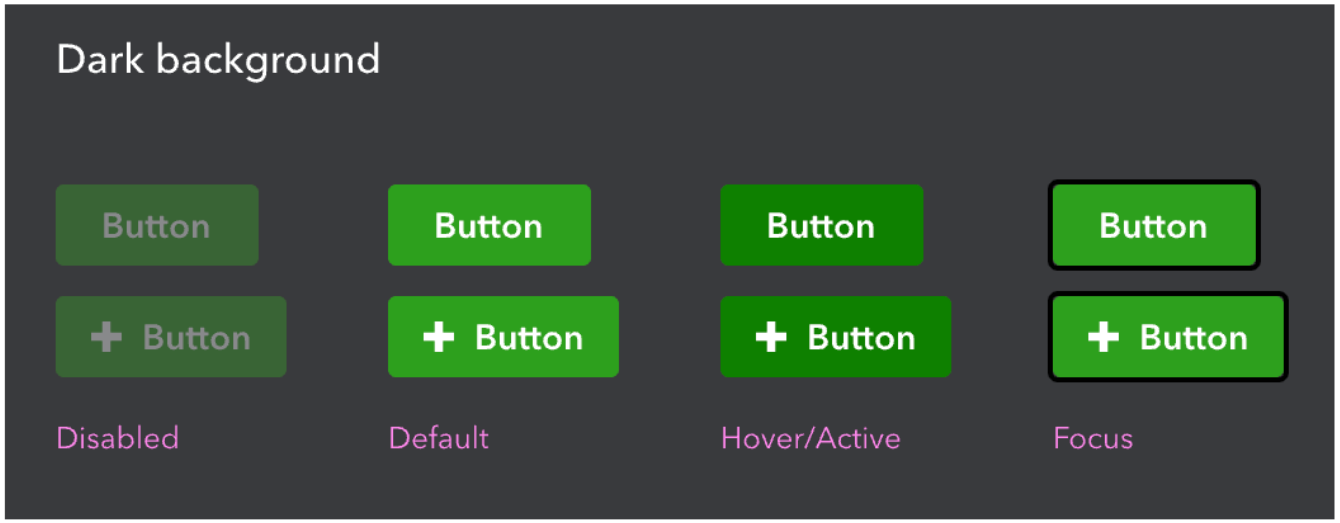
Material Design

· Material is a design syster

Primary-style buttons

- Use the primary button style for the most prominent call to action on the page.
- These buttons guide users to and through the main happy path in a task flow.







Material Design

- Material is a design system

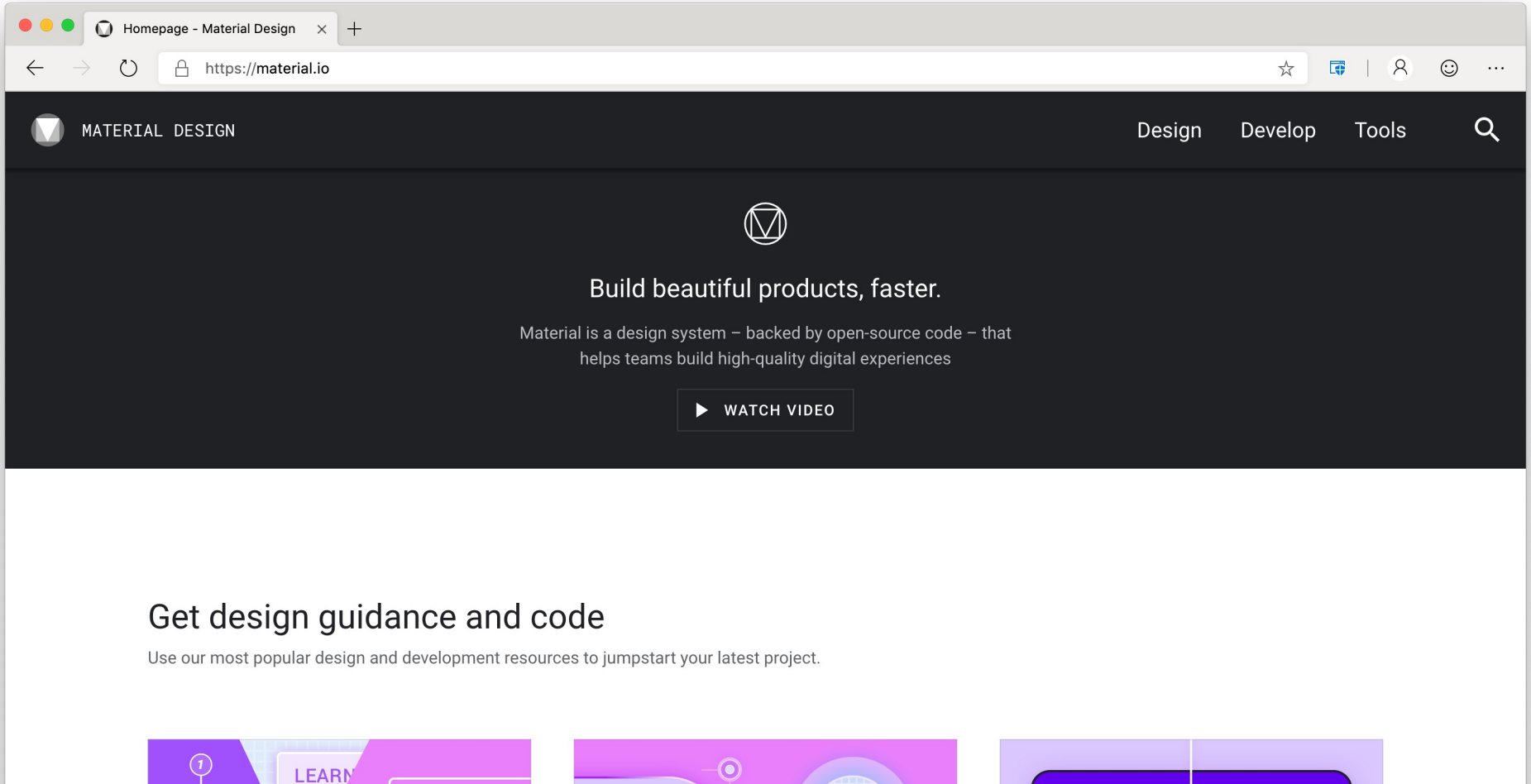


Material Design

- Material is a design system

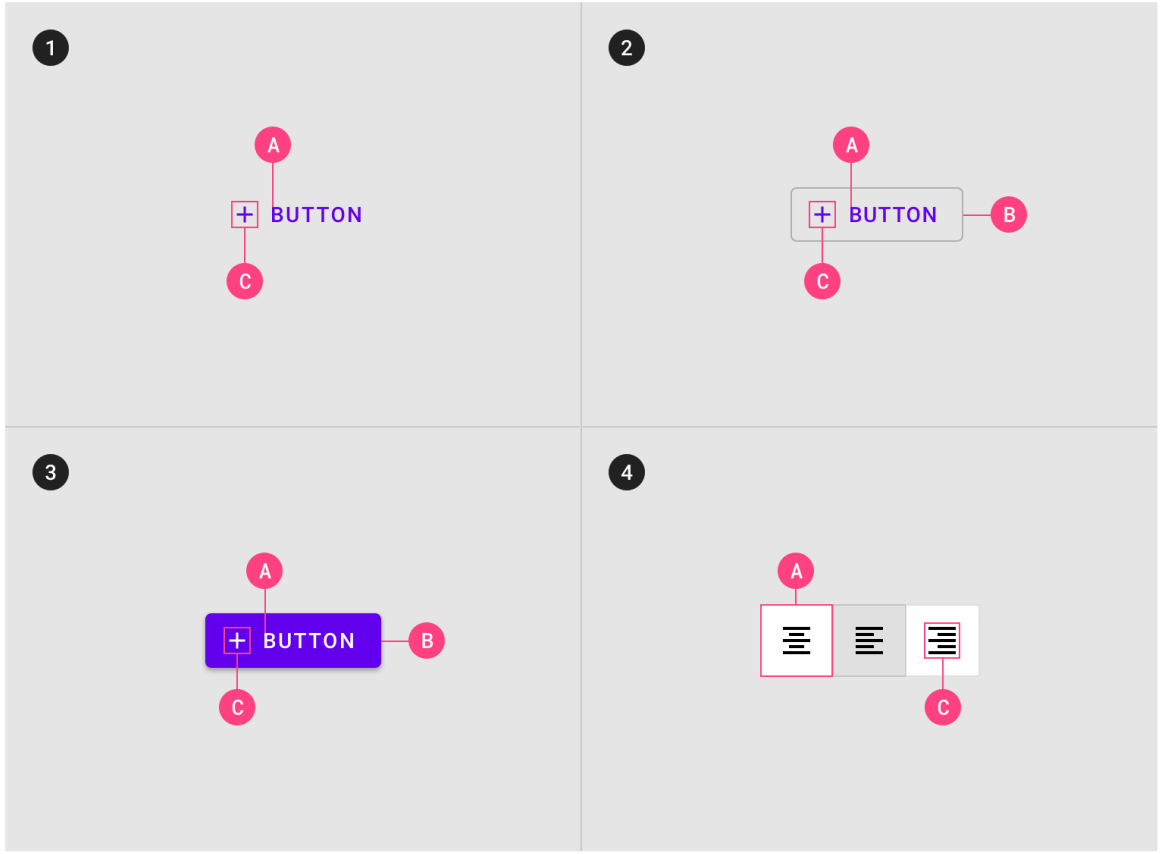
Material Design

- Material is a design system



Material Design

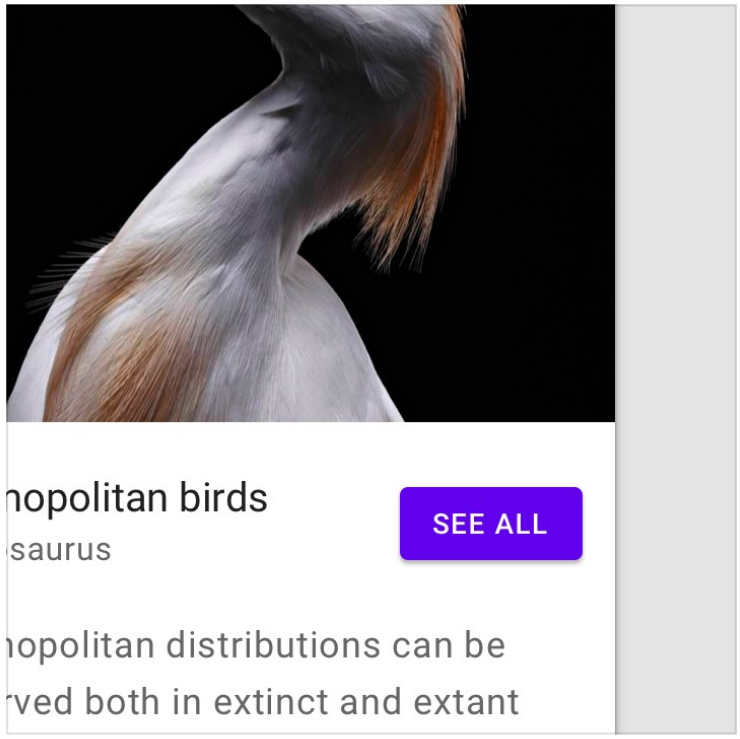
- Material is a design system



1. Text button
   A. Text label
   C. Icon (optional)
2. Outlined button
   A. Text label
   B. Container
   C. Icon (optional)
3. Contained button
   A. Text label
   B. Container
   C. Icon (optional)
4. Toggle button
   A. Container
   C. Icon

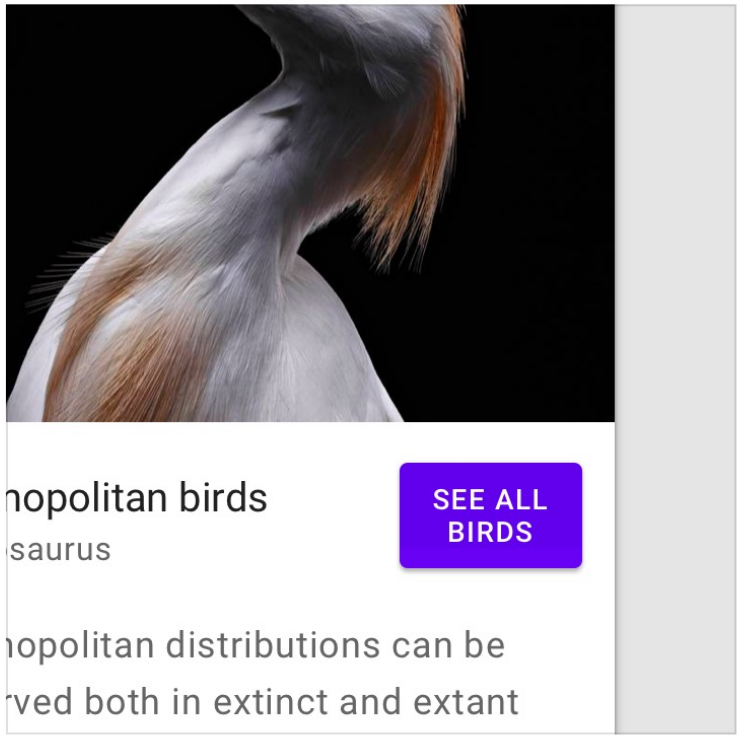
Material Design

- Material is a design system



Do.

Use capitalization, for languages that allow capitalization.



Don't.

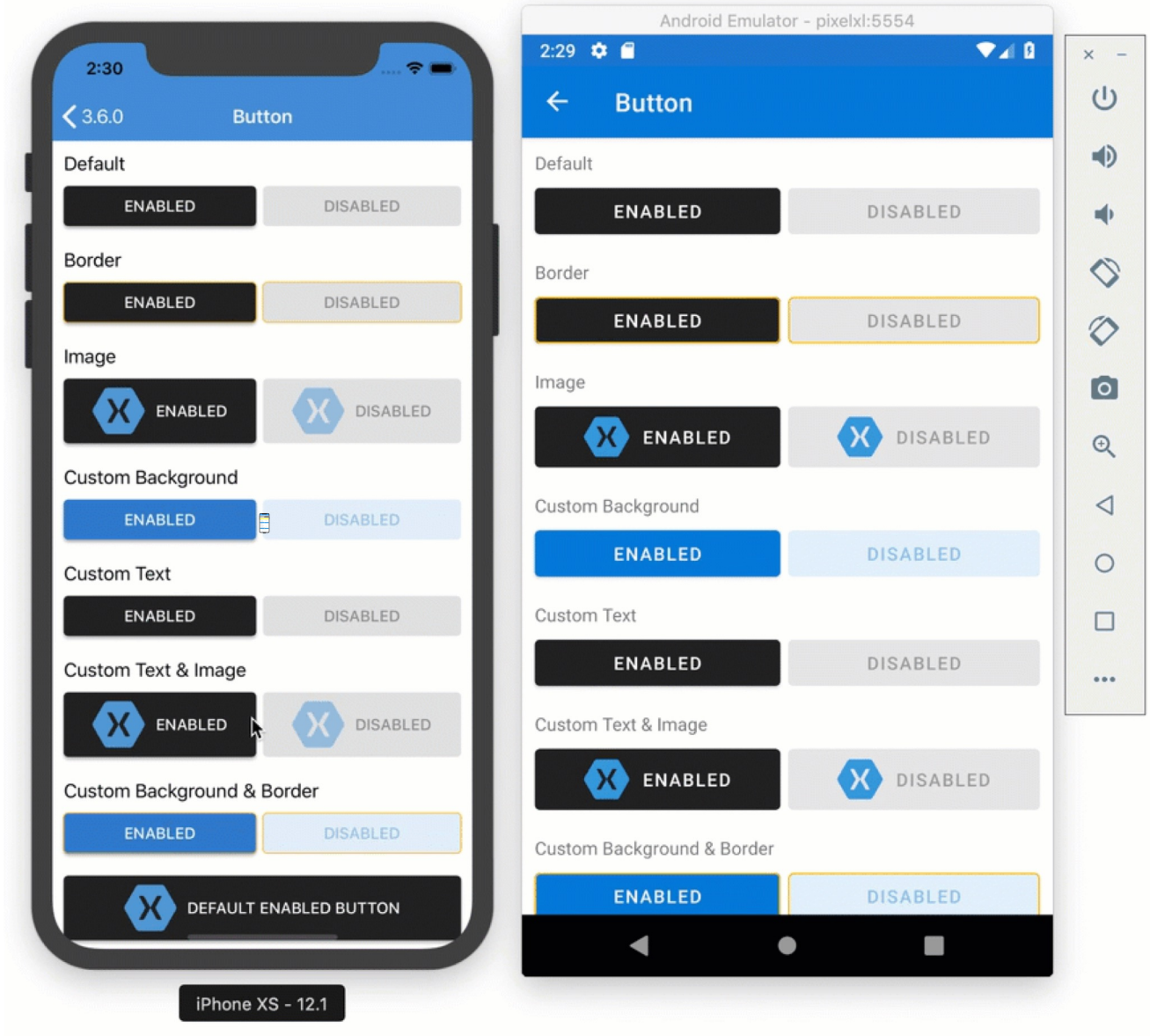
Don't wrap text. For maximum legibility, a text label should remain on a single line.

Visual="Material"

Material for iOS Controls

- ActivityIndicator
- Button
- Editor
- Entry
- Frame
- ProgressBar
- DatePicker
- TimePicker
- Picker
- Slider
- Stepper

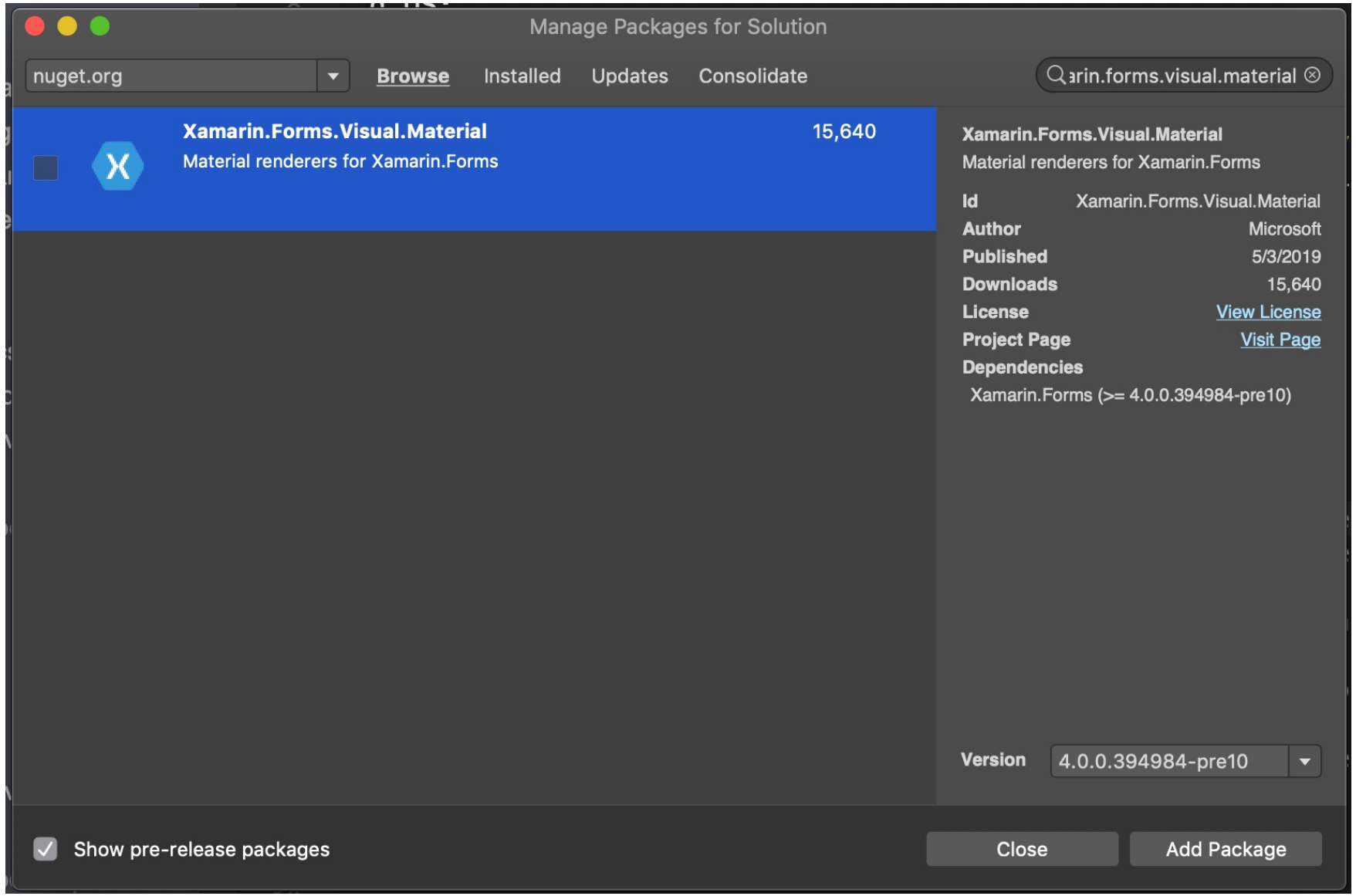
Material for iOS Controls



Material on iOS Demo


FUNNYCUTECI

1. Install the NuGet in Platform Projects



2. Initialize the Library (iOS)

```
global::Xamarin.Forms.Forms.Init();
global::Xamarin.Forms.FormsMaterial.Init();
```

2. Initialize the Library (Android)

```
global::Xamarin.Forms.Forms.Init(this, savedInstanceState);
global::Xamarin.Forms.FormsMaterial.Init(this, savedInstanceState);
```

3. Set Visual

```
<Shell Visual="Material">
```

3. Set Visual

```
<ContentPage Visual="Material">
```

3. Set Visual

```
<Button Visual="Material">
```

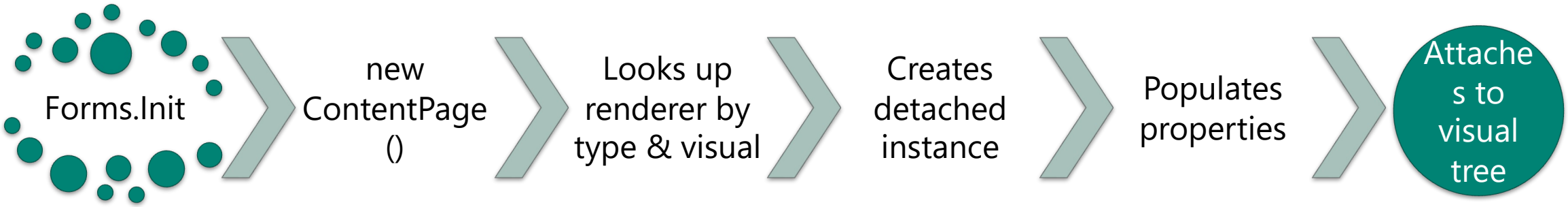
3. Set Visual

```
<Button Visual="Material" />
<Button Visual="Default" />
```


USEHIDDLES

Your Design System, Your Visual

Visual – Understanding The Renderer Process



Your Visual Definition

```
namespace Tailwind
{
	public class Tailwind.Visual : IVisual
	{
	}
}
```

Your Android Renderer

```
[assembly: ExportRenderer(typeof(Xamarin.Forms.Button),
    typeof(TailwindButtonRenderer),
    new[] { typeof(Tailwind.TailwindVisual) })
]

namespace Tailwind.Droid
{
      public class TailwindButtonRenderer : MButton, …
      {
```

Your Android Renderer

```
[assembly: ExportRenderer(typeof(Xamarin.Forms.Button),
    typeof(TailwindButtonRenderer),
    new[] { typeof(Tailwind.TailwindVisual) })
]

namespace Tailwind.Droid
{
      public class TailwindButtonRenderer : MButton, …
      {
```

Your Android Renderer

```
[assembly: ExportRenderer(typeof(Xamarin.Forms.Button),
    typeof(TailwindButtonRenderer),
    new[] { typeof(Tailwind.TailwindVisual) })
]

namespace Tailwind.Droid
{
      public class TailwindButtonRenderer : MButton, …
      {
```

Your iOS Renderer

```
[assembly: ExportRenderer(typeof(Xamarin.Forms.Button),
    typeof(TailwindButtonRenderer),
    new[] { typeof(Tailwind.TailwindVisual) })
]

namespace Tailwind.iOS
{
      public class TailwindButtonRenderer : ViewRenderer<Button, MButton>,
IImageVisualElementRenderer, IButtonLayoutRenderer
      {
```

Your Library Project

```
namespace Tailwind.Forms
{
    public static class Tailwind
    {
        public static void Init(Context context, Bundle bundle)
        {
                    // my only purpose is to exist so when called
                    // this dll doesn't get removed
        }
    }
}
```

Initialize the Library (iOS)

```
global::Xamarin.Forms.Forms.Init();
global::Tailwind.Forms.Tailwind.Init();
```

Set Visual

```
<Shell Visual="Tailwind">
```

Details

- Visual with Material available in Xamarin.Forms 3.6+
- Android Requirements
  - Target Framework 9.0 (API 28)
  - Android Support Libraries 28.0.0+
  - Minimum target API 21
  - Use an AppCompat theme or inherit from Material Theme
- Material Components for iOS
  - Updates coming.
- Visual API is available on ALL platforms that Xamarin.Forms targets



Docs and Demo

- https://docs.microsoft.com/en-us/xamarin/xamarin-forms/user-interface/visual/material-visual
- https://developer.xamarin.com/samples/xamarin-forms/UserInterface/VisualDemos/

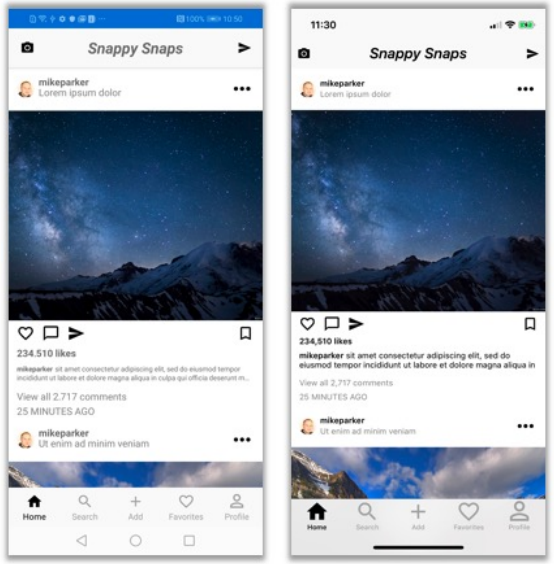


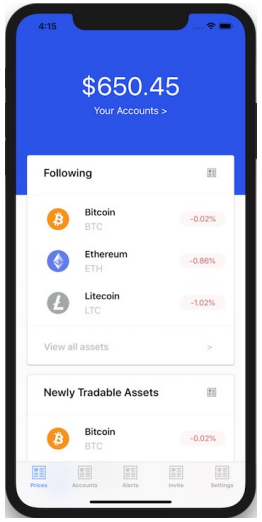
What's Next

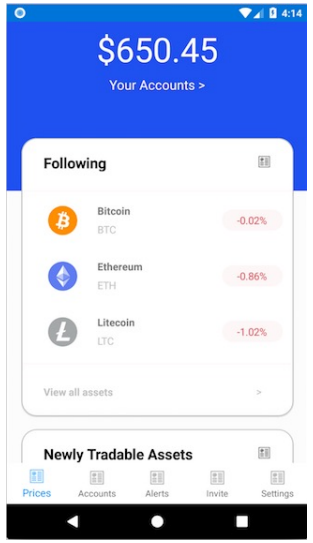
- Check out the Visual Challenge
  - https://github.com/davidortinau/VisualChallenge/pulls
  - https://devblogs.microsoft.com/xamarin/visual-challenge-conquered/
- Surfacing Visual properties

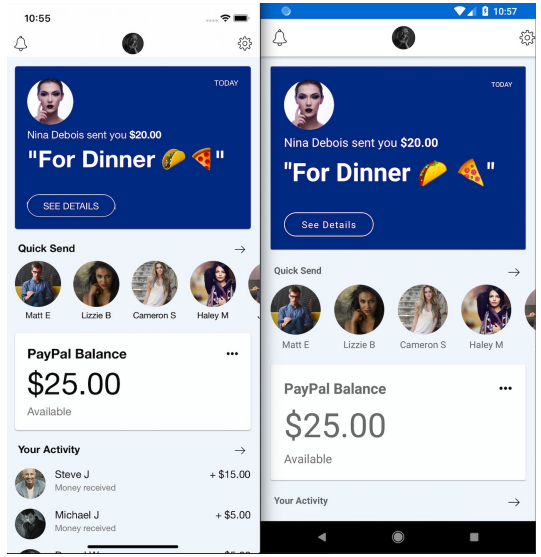


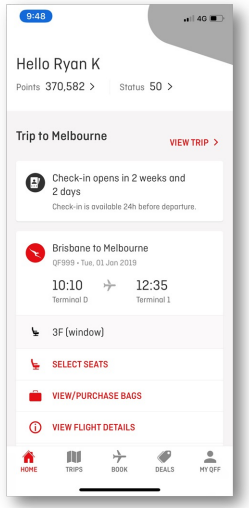
Go Forth and Build Apps Like



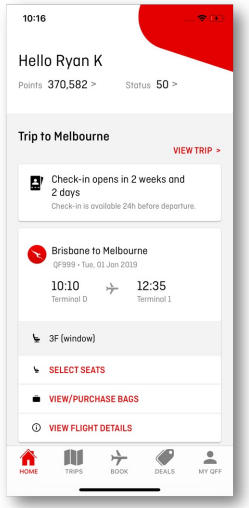




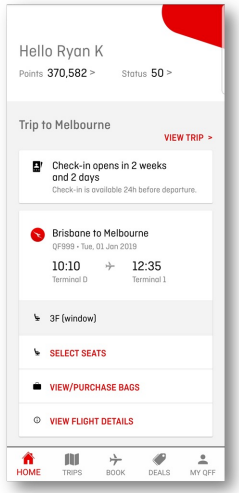




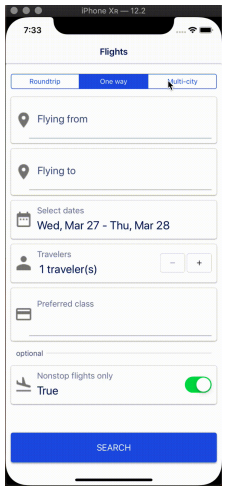
Source (iOS iPhone X)

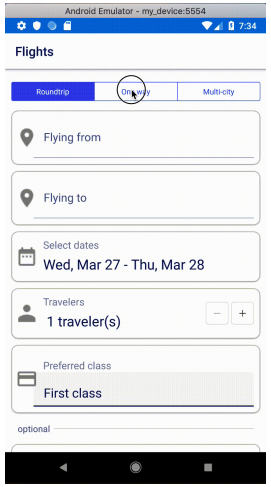


Visual (iOS iPhone X)



Visual (Android S9)






Have I made you proud?

Thanks!

David Ortinau
Senior Program Manager
Mobile Developer Tools
Xamarin

@davidortinau

Special thanks to my unofficial co-presenter Loki.

davidortinau



Microsoft

Questions?

Stop by the Xamarin booth in the hub
for stickers, pens, and more!

Microsoft

© Copyright Microsoft Corporation. All rights reserved.